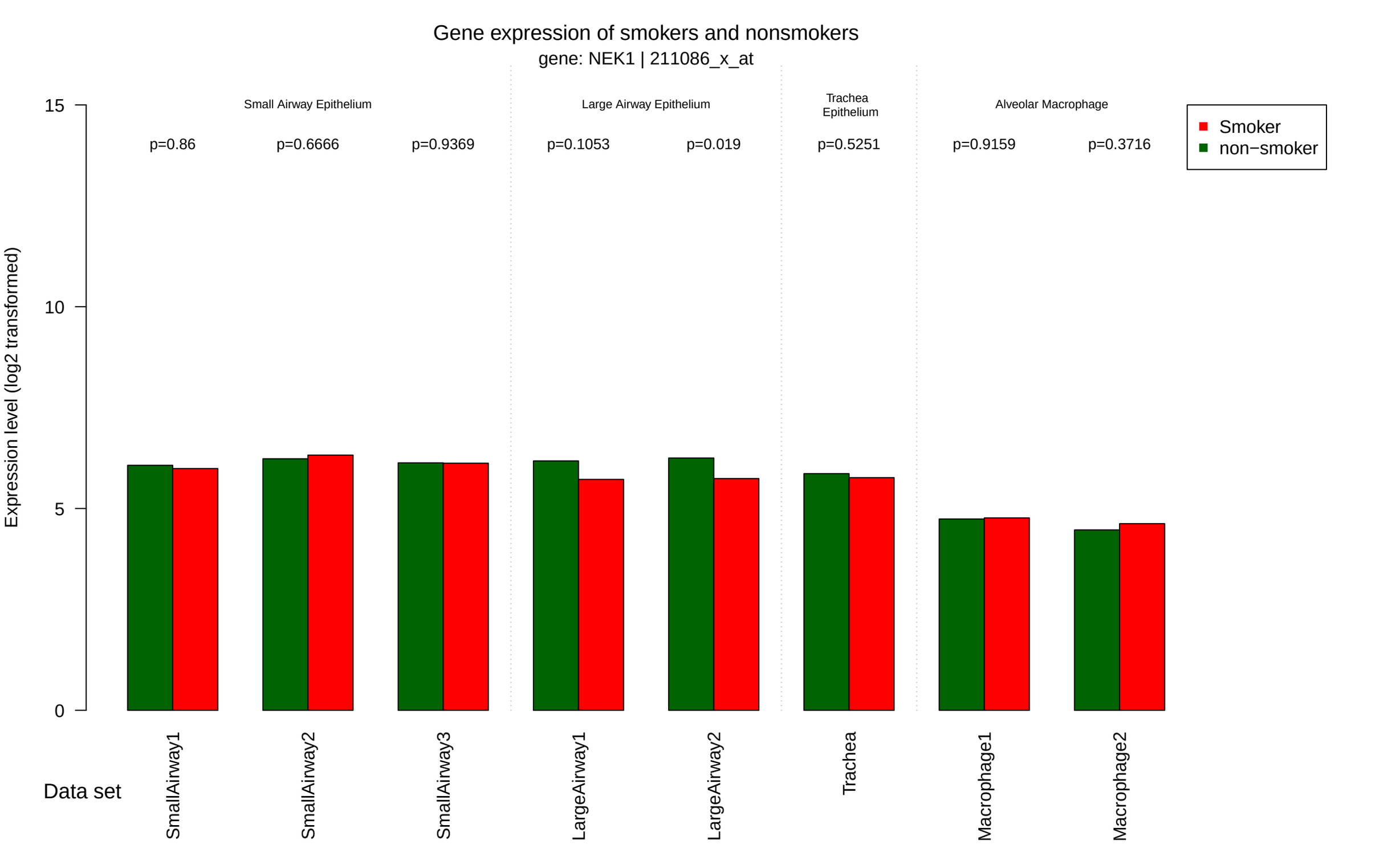
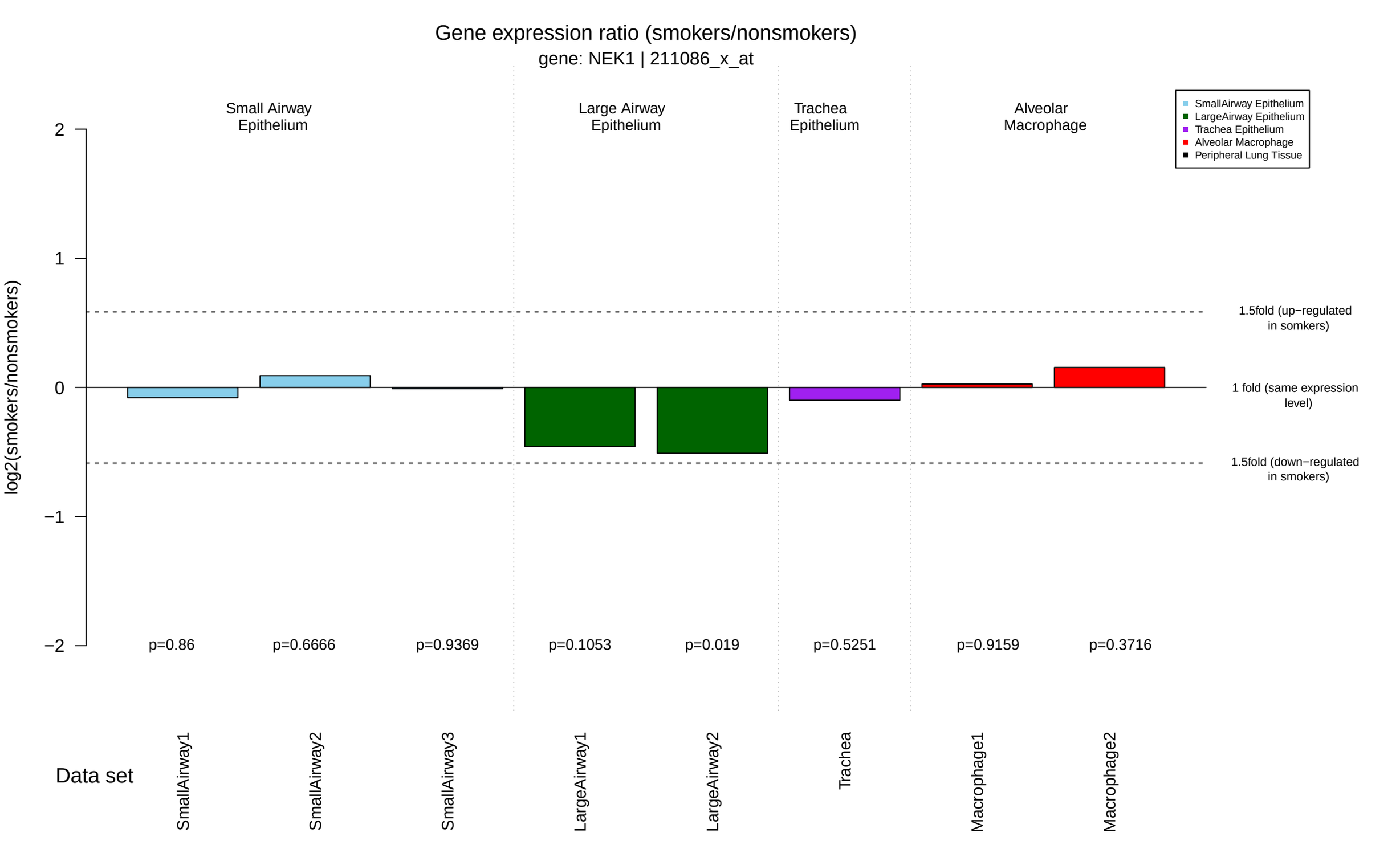
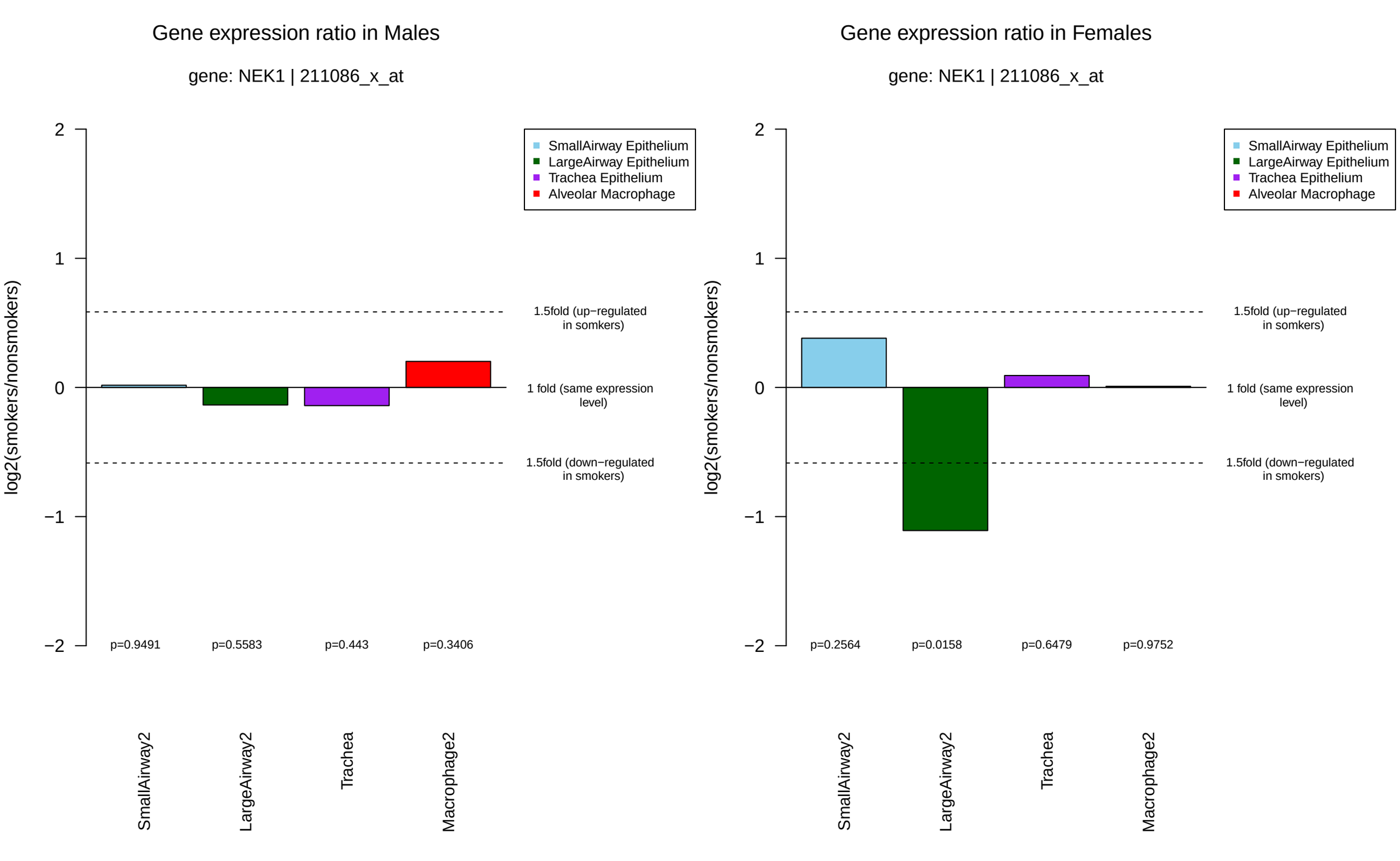
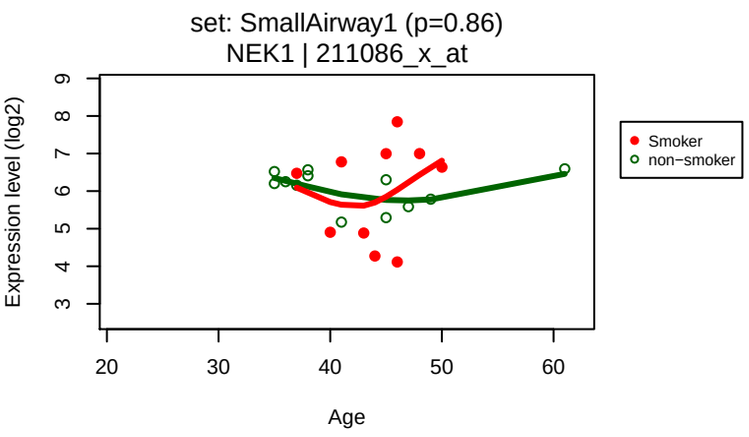
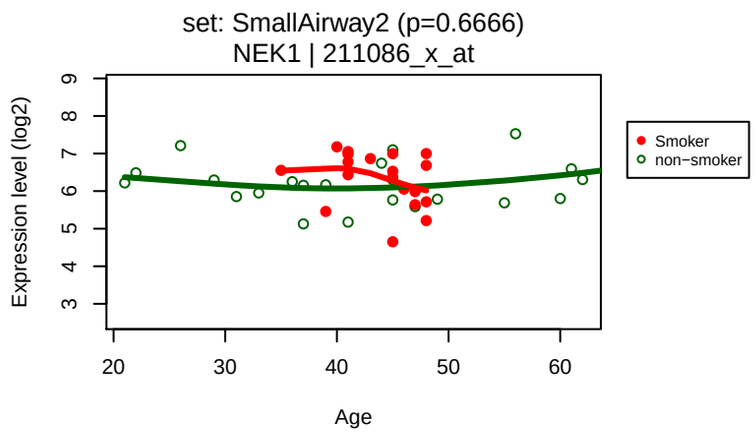
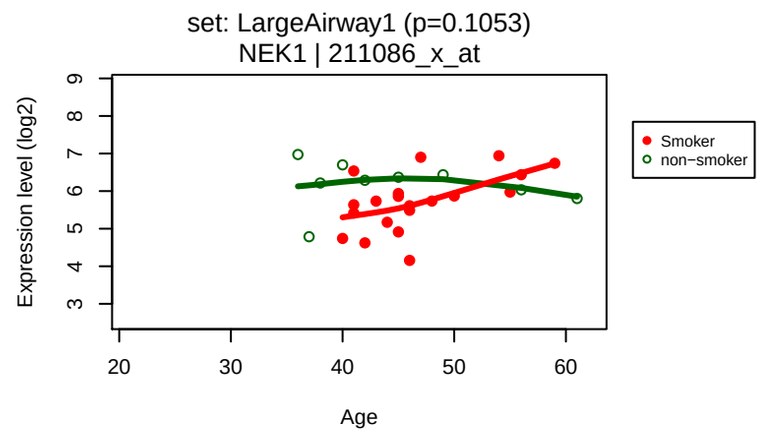

Gene expression of smokers and nonsmokers
gene: NEK1 | 211086_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.86
p=0.6666
p=0.9369
p=0.1053
p=0.019
p=0.5251
p=0.9159
p=0.3716
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

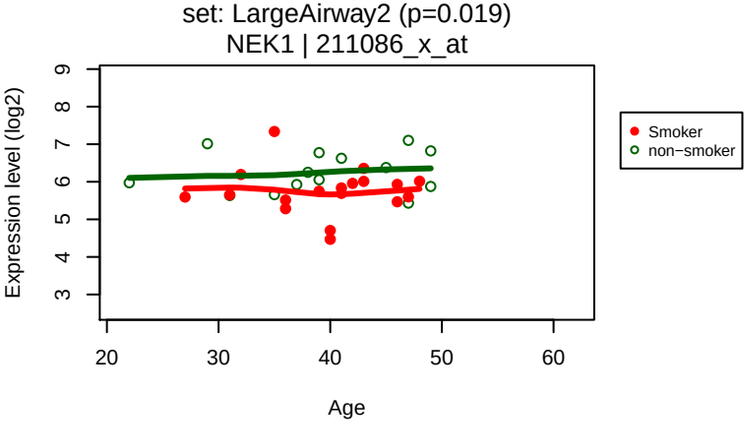

Gene expression ratio (smokers/nonsmokers)
gene: NEK1 | 211086_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.86
p=0.6666
p=0.9369
p=0.1053
p=0.019
p=0.5251
p=0.9159
p=0.3716
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

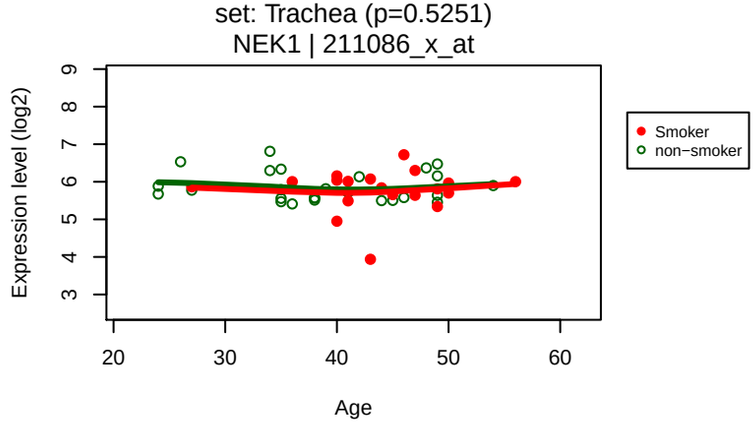

Gene expression ratio in Males
gene: NEK1 | 211086_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9491
p=0.5583
p=0.443
p=0.3406
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: NEK1 | 211086_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2564
p=0.0158
p=0.6479
p=0.9752
SmallAirway2
LargeAirway2
Trachea
Macrophage2

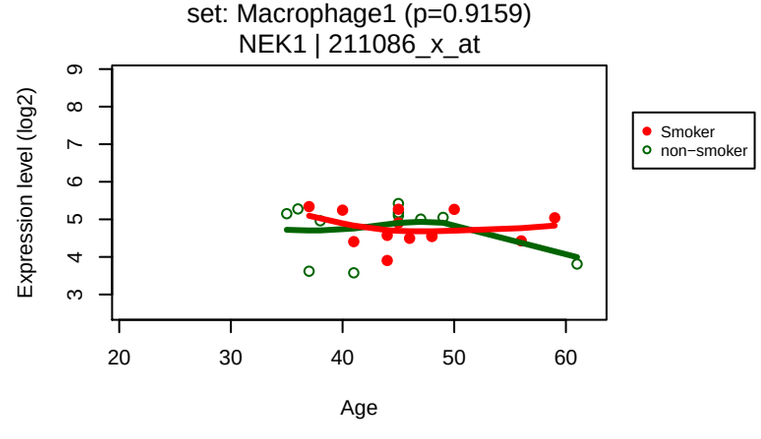

set: SmallAirway1 (p=0.86)
NEK1 | 211086_x_at
Expression level (log2)
Age
Smoker
non−smoker

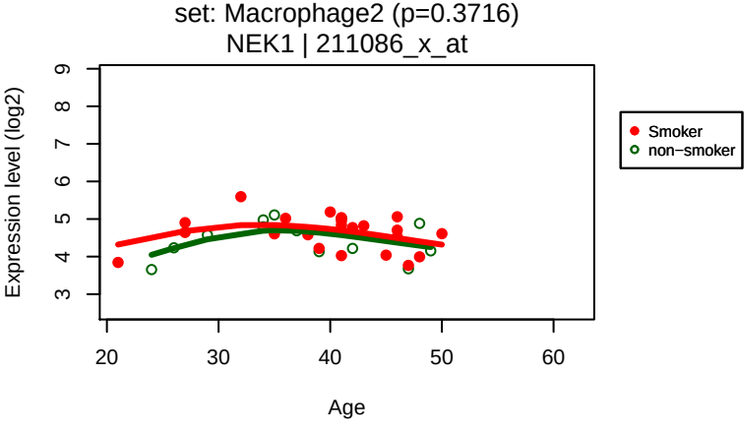

set: SmallAirway2 (p=0.6666)
NEK1 | 211086_x_at
Expression level (log2)
Age
Smoker
non−smoker

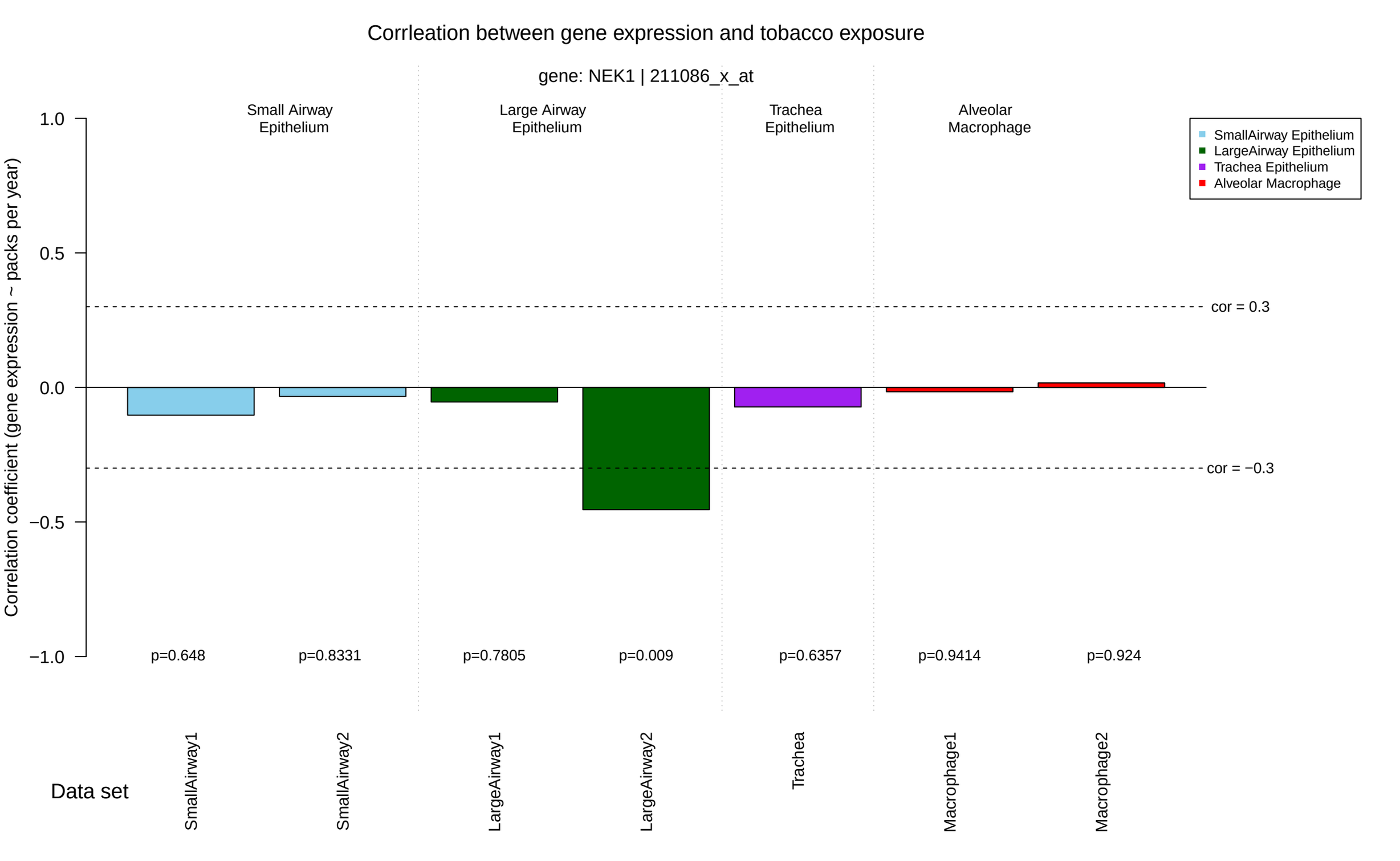

set: LargeAirway1 (p=0.1053)
NEK1 | 211086_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.019)
NEK1 | 211086_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5251)
NEK1 | 211086_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9159)
NEK1 | 211086_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3716)
NEK1 | 211086_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: NEK1 | 211086_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.648
p=0.8331
p=0.7805
p=0.009
p=0.6357
p=0.9414
p=0.924
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2